Problem Set 4

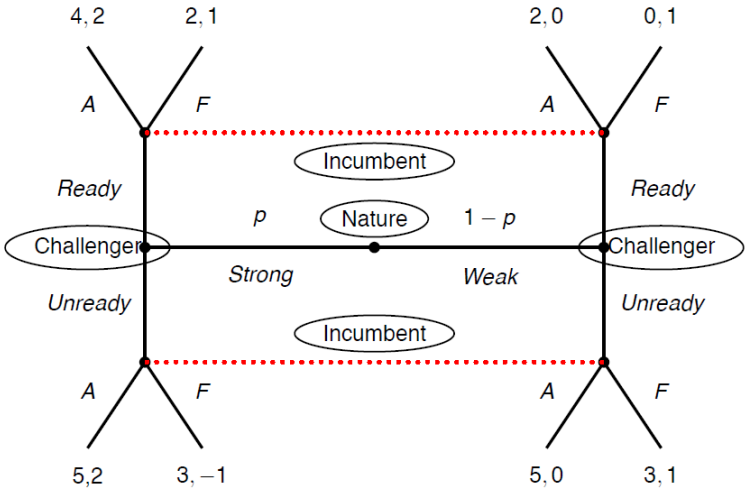
POLS W4732
Fall 2026

Due November 6 before the TA section

Instructions. For each formal claim, include two or three algebraic inequalities or best-response comparisons that establish it. Unless directly following notation from slides or the text of the problem, define your notation.

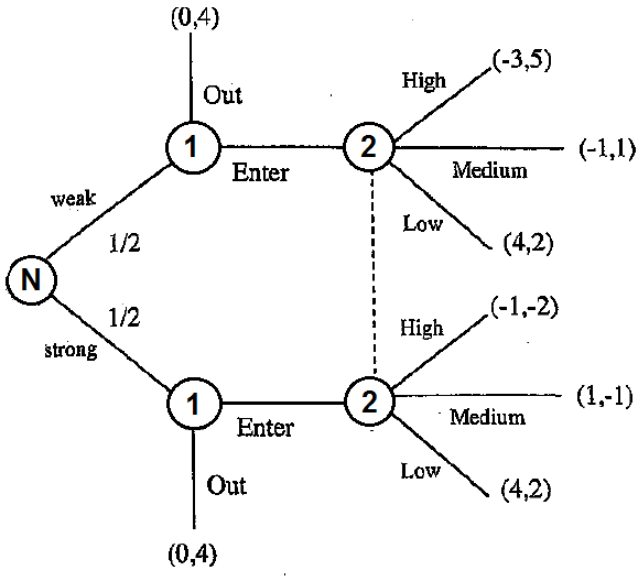
Artificial-intelligence policy. Students may not use generative-AI systems (e.g., ChatGPT, Claude, Gemini, Copilot, etc.) to formulate, derive, solve, explain, or write answers to this problem set. Course materials, ordinary calculators, and collaboration within the registered problem-set group are permitted. **Every group member is responsible for understanding and being able to explain the entire submission.**

Exercise 1. *Consider the crisis bargaining game:*



Show that if one modifies one payoff to one player in one terminal node of the crisis bargaining game, the separating equilibrium no longer exists

Exercise 2. *Consider the following game with an informed entrant, where $a_1(W)$ denotes the pure strategy of a Weak type of Pl. 1, $a_1(S)$ the pure strategy of a Strong type Pl. 1. Pl. 2 only observes an action a_1 and chooses her best response $a_2(a_1)$.*



1. *Describe the set of terminal histories and the set of non-terminal histories of this game*

2. *For each player, describe the information sets of this game*

3. *Does the profile ($a_1(S) = a_1(W) = Out$, $a_2 = High$) constitute a Perfect Bayesian Equilibrium?*

4. *Does the profile ($a_1(S) = a_1(W) = Enter$, $a_2 = Low$) constitute a Perfect Bayesian Equilibrium?*

5. *Does the profile ($a_1(S) = Enter$, $a_1(W) = Out$, $a_2 = Low$) constitute a Perfect Bayesian Equilibrium?*

Exercise 3. *There are two types of worker. Type θ has a marginal value product of $v_\theta = \theta$, where $\theta \in \{1, 2\}$. The cost of sending signal z for the low type is $C_1(z) = z$ and for the high type is $C_2(z) = z(1 - \epsilon)$. With probability π, the worker is a low type.*
The equilibrium wage is going to be the expected marginal value product of the worker.[1]

1. *Show that there is a continuum of PBEs in which Type 1 chooses $z = 0$ and Type 2 chooses signal $z \in Z \equiv (1, 1/(1 - \epsilon))$*

2. *Consider now the possibility of a pooling equilibrium. What are the possible values of z compatible with a pooling equilibrium?*

3. *What is the highest equilibrium payoff to each type of worker from a pooling equilibrium? Can we conclude that Type 2 is always better off under a separating equilibrium?*

[1] One way of justifying this is to assume that there are two firms competing á la Bertrand for the worker.

Exercise 4. *Consider the baseline version of the cheap talk model. Prove that the Sender is worse off in a babbling equilibrium than in a partially revealing equilibrium with two possible reports.*